

Try Making This TFT-Based GPS Analogue Clock Using Arduino

This GPS-based project displays analogue clock and room temperature on a TFT display. GPS time is a precise time standard that is related to Coordinated Universal Time (UTC). The displays of most electronic hobby projects are based on LCD, GLCD, OLED or TFT. This project is based on an 8.89cm (3.5-inch) TFT display having 480 × 360 pixel resolution (Fig. 1).

The latest Chinese TFT displays are quite low in price but work perfectly with Arduino and Raspberry Pi. There are two different types of TFT shields available: one with 26 pins (13 × 2 DIL) for Raspberry Pi and the other as Arduino TFT shield for Arduino Uno board.

Arduino TFT shield

Arduino TFT shield can be perfectly mounted on top of an Arduino Uno board. But a major disadvantage is that after mounting the shield on Arduino

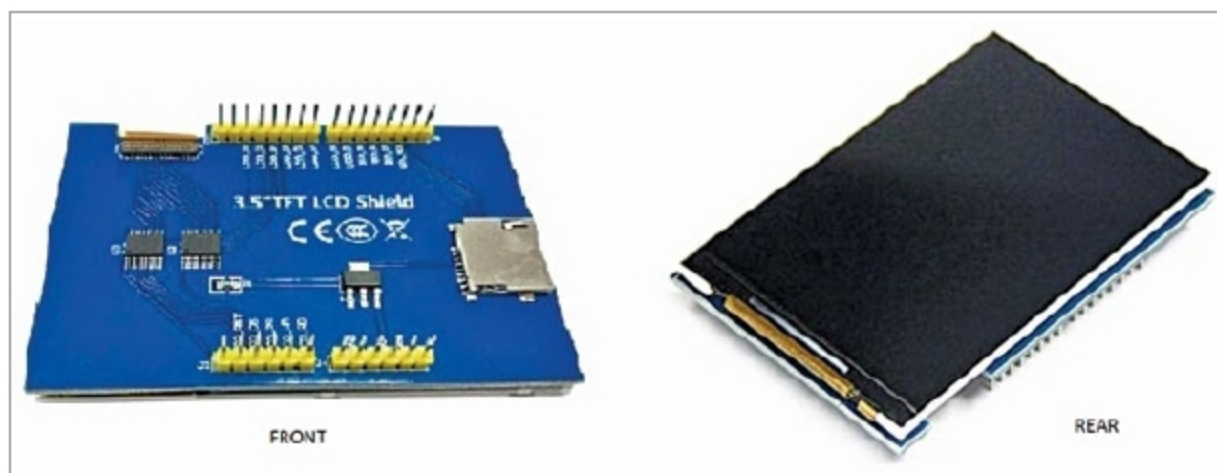


Fig. 1: TFT LCD Arduino shield

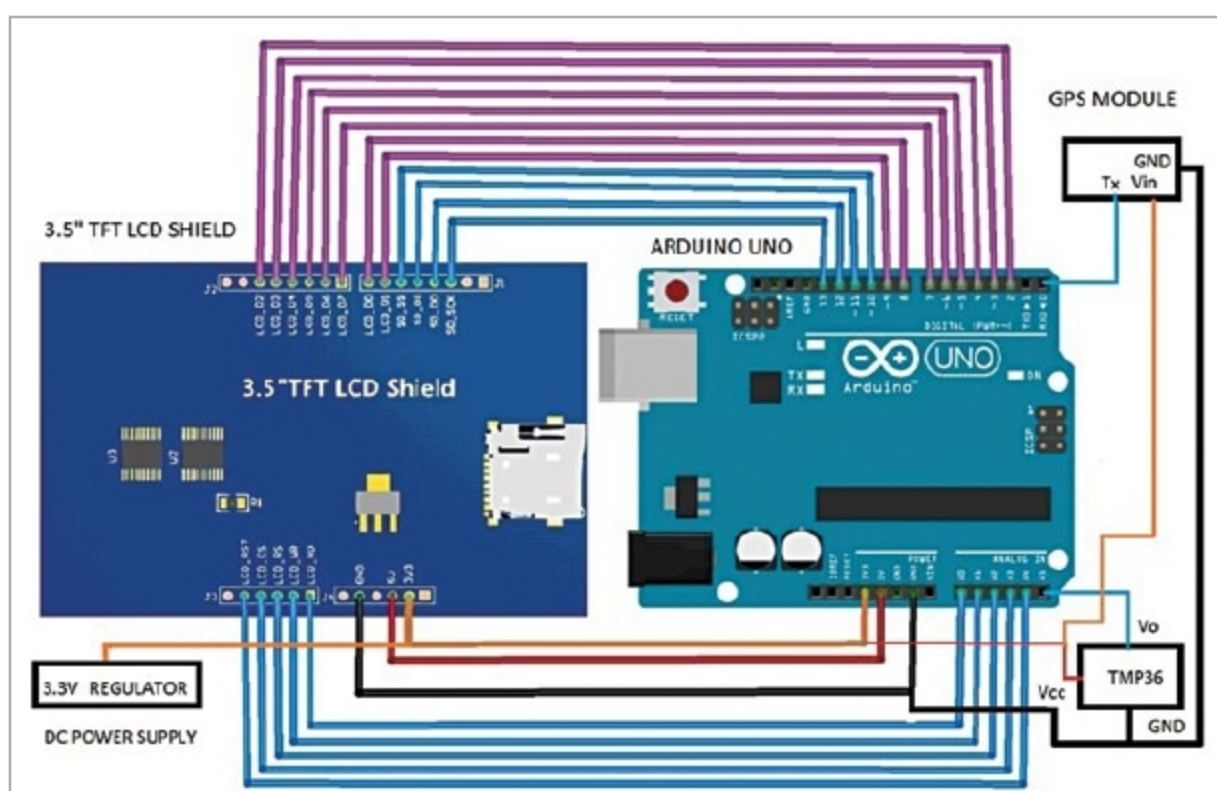


Fig. 2: Circuit diagram of the GPS analogue clock

Uno board, it is difficult to use its GPIO pins for any other application.

The Arduino TFT shield has a micro SD card attached that gets connected to its serial peripheral interface (SPI) pins for communication with the microcontroller. This micro SD card is not used in this project.

The main hurdle with this display is that it is not common and is available only from mcufriend.com. Fortunately, the `mcufriend_kbd.h` header file is freely available and it works out of the box for this and many other

similar displays. The other header file required for this display is `Adafruit_GFX.h`. Both these header files, along with the main source code, are available on EFY website.

This project creates an analogue dial clock along with digital date, time and temperature display using an LM35/TMP36 temperature sensor. The time signal is derived from a U-Blox NEO-6M GPS receiver module. Once the technique is understood, it can be deployed for many other applications.

Pin-to-Pin Connections Between TFT Shield and Arduino Uno

Arduino TFT shield pin	Arduino pin
RD	A0
WR	A1
RS	A2
CS	A3
RST	A4
D0	8
D1	9
D2	2
D3	3
D4	4
D5	5
D6	6
D7	7
3.3 volts	3.3 volts
Gnd	Gnd
5 volts	5 volts